



ONEAQUAHEALTH IEEE GLOBAL HACKATHON 2026 · TRACK 7 DIGITAL HEALTH STANDARDS

StreamChart

Every patient has a chart. Now every stream does.

A 60-second citizen check-up, a sealed AI second look, confidence grades, a care plan signed by an ecologist, and a reply. In FHIR R4.

Live app: streamchart-oneaquahealth.netlify.app

Code: github.com/chessoren/streamchart

Submission fields

Project name

StreamChart

Elevator pitch

Every patient has a chart. Now every stream does: a 60-second citizen check-up, a sealed AI second look, confidence grades, a care plan signed by an ecologist, and a reply. In FHIR R4.

Track

Track 7 — Digital Health Standards (also covers Tracks 3, 2, 5, 4 and 6).

Built with

react typescript vite gemini google-ai-studio gemini-vision structured-output
hl7-fhir fhir-r4 hapi-fhir open-meteo sha-256 commit-reveal css3 html5 lucide
qrcode playwright vitest netlify github fish-audio ffmpeg wikimedia-commons
claude-code

Try it out

Live app — <https://streamchart-oneaquahealth.netlify.app>

Source code — <https://github.com/chessoren/streamchart>

Emergency scene — <https://streamchart-oneaquahealth.netlify.app/#/simulation>

A stream on the public HAPI FHIR R4 server — <https://hapi.fhir.org/baseR4/Patient/81129>

Demo video — StreamChart-demo.mp4 (4 min 24 s, English subtitles)

About the project

Every patient has a chart. Now every stream does.

Track 7, Digital Health Standards (primary). The project also covers Track 3 (AI-supported assessment), Track 2 (data-to-insight), Track 5 (community and engagement), Track 4 (storytelling) and Track 6 (early warning).

Inspiration

There is a stream at the bottom of a path that thousands of people walk past every week. They see it without looking at it. In August a white foam gathers against a stone, and nobody says anything. The foam is not hidden, people just assume someone else has already noticed. When somebody finally does say something, the observation goes into a form, the form goes into a database, and the person who took the trouble never hears about it again. The next time they see foam, they don't bother.

That is the quiet failure at the heart of urban citizen science. It isn't a lack of goodwill. When the OneAquaHealth project asked for help, people downloaded the app, walked to their streams and answered the questions. The failure comes after: nothing comes back. The hackathon's Track 5 names it plainly as *low repeat engagement*. Track 3 names the other side of the same coin: *citizen observations can be inconsistent and error-prone*, so authorities hesitate to act on them. Volunteers stop contributing because nothing visible happens, and nothing happens partly because nobody trusts the data. Each problem keeps the other going.

When I read through what OneAquaHealth has built over the last few years, I was struck by how complete it already is. It has a citizen science app with standardised indicators, dashboards for five research cities (Benevento, Coimbra, Ghent, Oslo, Toulouse), a resilience map crossing biodiversity, weather, pathogens and satellite data, the GEOSSIP data-sharing platform, a decision-support system that goes from indicators to likely alterations to stressors to restoration measures, DipteraCAST for forecasting mosquitoes and other dipterans, and a carefully written Catalogue of Measures. In medical terms, the project has built every organ of a health system for urban streams.

What it hasn't built is the patient.

Every one of those tools answers a useful question: what do we see, what do we predict, what should we do. None of them answers the question a resident actually asks: *what is happening to my stream, and did what I did make any difference?*

Medicine faced exactly this problem more than a century ago, and solved it with three inventions we now take for granted. The **medical record** keeps the story of one patient over time. **Levels of evidence** say how far each piece of information can be trusted. The **care loop** connects a symptom to a diagnosis, to a prescription, to a follow-up visit, and finally to recovery. A modern hospital isn't impressive because of any one machine. It works because every machine writes into the same chart, and every chart has a doctor who signs it.

An urban stream is a patient that has never had a chart, a doctor or a follow-up visit. So the idea was simple, maybe obviously so once said out loud: **give every stream a medical record, and run citizen science as a care loop**. The hackathon's tagline is *from streams to systems*. StreamChart adds the missing link in the middle: *from streams, to patients, to systems*.

I come from theatre. In theatre you learn quickly that an audience doesn't remember features, it remembers a character. I wanted the stream to become one: a patient with a name, a state, a history and even a voice, without ever losing the scientific rigour that the people who built OneAquaHealth have every right to expect. The emotion I aimed for is what I'd call *serious tenderness*. The tenderness comes from a stream that says "nobody has looked at me for nineteen days". The seriousness comes from confidence grades, co-signatures and provenance on every fact.

What it does

StreamChart is a working web application, live at <https://streamchart-oneaquahealth.netlify.app>, built around one consistent translation from medicine to streams:

Medicine	StreamChart
Patient	A stream reach, short enough to be described in one look
Medical record	The stream's chart: check-ups, photos, hypotheses, care, outcomes
Consultation	A 60-second check-up on the bank
Symptom	An abnormal observation: foam, colour, odour, stagnant water, eroding banks
Vital signs	Background measures the project already collects (temperature, water quality, weather)
Laboratory test	Specialised analyses, such as a DipteraCAST prediction or an expert identification
Diagnosis	A <i>hypothesis</i> , always with a confidence level
Level of evidence	A confidence grade from A to D on every fact
Prescription	A care plan made of measures from the OneAquaHealth catalogue
Doctor's signature	Co-signature by a referent ecologist
Follow-up visit	A follow-up check-up from the same viewpoint
Recovery	Remission
Care team	The residents, schools and associations that follow the stream

Four rules keep the metaphor honest, and the code enforces all of them. We say *hypothesis*, never *diagnosis*. Nothing is *validated* until a competent human has signed it. We never dramatise: a stream under watch is not dying. And we never accuse anyone: the record describes states and plausible causes, never culprits.

Each stream is always in one of four states, visible on the map and at the top of its chart: **not followed** (nobody has looked for more than two weeks, which is the saddest state of all), **stable**, **under watch** (one or more signs deserve a second look) and **alert** (several concordant signs, plus a risk for the environment or for people). There is also a fifth moment, which is a celebration rather than a state: **remission**, when a stream that was in alert comes back to stable after care.

The loop has five steps.

1. Triage: the 60-second check-up, with a sealed AI second look

A resident scans the sign at the foot of the stream, or opens the app. The stream introduces itself, and if nobody has visited for a while, it says so: *"Nobody has looked at me for nineteen days."* A safety reminder comes first: observe from the bank, never step into fast or deep water, children come with an adult.

The resident takes one photo and answers seven questions with very large buttons. The questions follow the OneAquaHealth citizen indicators that can be judged with the naked eye and no equipment: water clarity and colour, foam, odour, flow, banks and bed, riparian vegetation, and litter. Each question comes in **three language levels**, chosen with a switch: *Child* ("the water looks like dirty milk"), *Curious* ("cloudy, with whitish deposits") and *Expert* ("elevated turbidity"). The underlying information is identical; only the vocabulary changes, and technical terms open a one-line explanation. This answers Track 1's request to simplify ecological terms without dumbing down the data.

Most AI applications show their answer immediately, and people accept it without thinking. At that point the observation is no longer human. StreamChart does the opposite, and this is the mechanism I'm proudest of. **The citizen answers first.** While they answer, Google's Gemini vision model looks at the same photo. It never sees the citizen's answers, and its own answer is **sealed**: hashed with SHA-256 the moment it arrives and kept hidden. The citizen then seals their check-up with a deliberately solemn button, *"I confirm my check-up"*, and a small stamp animation. Their answers are hashed too. Only then is the AI's answer revealed. It is a commit-reveal protocol, the same idea used in sealed-bid auctions and blockchain voting, applied to a citizen and an AI looking at the same stream.

The reveal is a comparison, not a correction. The two answers sit side by side, field by field. Agreements get a discreet green mark. Disagreements are highlighted, and the AI's observations are drawn as boxes on the photo with a plain sentence, such as *"Here: man-made concrete retaining wall framing the water body."* For each disagreement the citizen chooses: **keep my answer**, **adopt the AI's**, or **ask for a second opinion**. The last word always belongs to the human.

The AI is allowed to say "I don't know", and it uses that right. It answers *unsure* whenever the photo doesn't let it judge, and it is hard-wired to answer *unsure* for odour, because a photo cannot carry a smell. Ecologists will notice that detail, and I think it is one of the things that shows we understand the science.

Finally, the citizen immediately sees what their check-up changed: the stream moves from *not followed* to *under watch*, or their check-up joins a neighbour's and raises its grade. Nobody leaves without a reply.

2. The record: the stream's chart

The chart is the screen that carries the whole idea. It reads like a hospital record with the warmth of a personal page:

- **Identity:** name, city, district, length of the reach, surroundings, date of the last visit, a photo, and a state badge, followed by one sentence in everyday language: *"Trois Ponts brook is not doing well: several signs agree. Three people have looked at it this week."*
- **The pulse:** a heartbeat-like trace drawn from the stream's state over four months, with a blip for each check-up. One glance tells you whether it is recovering, relapsing or asleep.
- **The life of the stream:** a timeline, most recent first, of check-ups (with the photo and how each was compared with the AI), background measures, hypotheses, the health indication, care plans and signatures, interventions, follow-up visits and remissions. Each entry carries its author and its confidence grade.
- **The care team:** small avatars with pseudonyms and roles that tell a story without ranking anyone: the *sentinel* who reported first, the *vigil* who comes back regularly, the *scout* whose eye is often closest to the consensus, *the class* that adopted the stream, and the referent ecologist.
- **Details for experts:** the current findings with their grades and the reasons for each grade, the grading rules, the seals of the latest check-ups, and the HL7 FHIR export.

The chart works on a phone, a ten-year-old can read it, and an ecologist can take it seriously. Anything that doesn't help understand the state, the history or the next action stays in the expert details.

3. Assessment: confidence grades, hypotheses and a seven-day health indication

StreamChart doesn't claim to tell the truth about a stream. It says **what can reasonably be thought, and how far that can be trusted**.

Every finding carries a **confidence grade**, modelled on levels of evidence in medicine:

- **A:** solid. There is a photo, the person and the AI agree, and others corroborate, or an expert has validated it.
- **B:** reliable. A photo with person-AI agreement, or two concordant people.
- **C:** plausible but isolated. One person, or an unresolved disagreement.
- **D:** fragile. No photo, imprecise, or contradicted.

The grade is a property of the information, not a judgement of the person. It moves over time. Mathis reports foam on his own: grade B (photo, plus agreement with the AI). His report of an odour is grade C, because only one nose was there and a photo can't confirm a smell. The next day Léa reports the same thing, and the odour climbs to B. When the referent ecologist signs the care plan, the findings she relied on reach A. Watching a grade rise is one of the most satisfying moments in the product: the community becomes more credible than the sum of its parts. Disagreements are never smoothed away. A contradiction within a week is shown and lowers the grade.

When several symptoms combine, the engine proposes one or two **hypotheses**, never a certainty. For example: *"Possible input of wastewater or polluted runoff — confidence medium."* Each hypothesis lists its reasons (persistent foam reported by two people, unusual odour, cloudy water), offers a *Why?* button that opens the underlying check-ups and their grades, and says what would confirm it: *"a visit by the referent ecologist, or a water-quality measurement upstream."* The wording always speaks of probability, never of blame.

Then comes the bridge from ecosystem to human health, the heart of One Health. The system crosses observed symptoms with a **live seven-day weather forecast from Open-Meteo**. If stagnant water has been reported and several hot days are forecast, the chart shows an *indication*: *"Conditions favourable to mosquito proliferation over the next 7 days."* The indication comes with its reasons, the forecast it used, a reassessment date, and the explicit note *"indication, not an epidemiological forecast."* The current rule is transparent and deliberately simple. The chart already has a slot, labelled as a laboratory analysis, ready to receive predictions from the project's own DipteraCAST tool.

4. Treatment: a care plan co-signed by a human

A diagnosis without treatment is a worry, not a system. When a stream goes into alert, the system proposes a **care plan** of at most five lines. StreamChart doesn't invent measures: each line is a measure from the **OneAquaHealth D2.4 Catalogue of measures for urban aquatic ecosystems rehabilitation**, quoted with its real section number and title (for example *§4.2.2 Sewer system and point-source improvements, §4.2.3 Litter, plastic, and hydrocarbon control, §4.1.2 Active riparian vegetation restoration*), its line in the catalogue's hierarchy (first line essential, second line structural, third line social, fourth line complementary), a plain-language gloss, the hypothesis it addresses, who can carry it out (the city, an association, residents, the ecologist), and its urgency and effort. Part of every plan is meant for residents themselves, such as regular check-ups or a bank clean-up, because a community that only observes eventually drifts away.

Until a competent human signs, the plan is marked **"proposed by the system."** Signing is the solemn moment of this step. The referent ecologist opens the file, looks at the photos, confirms or questions the hypothesis, removes any line she judges premature (with a reason that stays in the record), and presses **"I validate this plan."** The band turns from *proposed* to *validated by Dr Ferreira, referent ecologist*, a stamp prints, and her name, role and date go into the record together with a SHA-256 signature hash. The system is never allowed to write "validated" on its own. Most AI projects talk about a human in the loop; StreamChart makes that human's signature the visible centre of the workflow. Sealing the check-up and signing the plan are the two solemn gestures at the two ends of the loop, and they share the same visual grammar.

For city services there is a **ward room**: every stream in the city sorted by urgency, each with a three-line summary, the average confidence grade, the measures proposed, and three buttons: *open the file, send to the ecologist, plan the work*. The technician doesn't need to understand ecology. They need to know where to go first.

5. Follow-up: the loop closes

This is the step almost nobody builds, and it is the one that makes people come back. Without it StreamChart would be a nice reporting tool. With it, it becomes a system people want to return to. The message is simple: *what you do has consequences, and you get to see them.*

Every time a check-up contributes to something, the person receives a short, personal, concrete message at a precise moment:

- when it is compared: *"Your check-up was compared with one other from the last three weeks. It is grade C."*
- when the stream changes state: *"Trois Ponts brook is now under watch. Your report is the first in the chain."*
- when a plan is validated: *"Your check-up on Wednesday triggered an inspection. Dr Ferreira validated the care plan. A follow-up visit is planned in three weeks. You are part of its care team."*
- when an intervention is done: *"The city completed 'Litter, plastic, and hydrocarbon control'. Would you take the photo again from the same spot?"*
- and when the stream recovers: *"Trois Ponts brook is in remission. Thank you for being the first to notice."*

There are no points, flames or streaks. The tone is that of a person, not a machine.

A **follow-up check-up** after an intervention closes the episode. If it confirms the improvement, the stream goes into **remission** and the chart shows a **recovery curve** with the three dates that matter (first report, plan signed, follow-up visit) and the names of the people involved, plus a **before/after slider** between the first photo and the follow-up photo.

The stream's voice

The stream has a voice, and that voice is the emotional signature of the project. A *Listen to the stream* button opens a short first-person account of three to six sentences, read on screen or aloud, in the same three registers. *"Sunday, Mathis saw my water turning cloudy and foam near my stone. Monday, Léa saw the same, and a faint smell and still water in my side arm. Three people saw it, so it's serious. Dr Ferreira signed a care plan for me. For the first time in a long while, I know someone is taking care of me."*

It follows one absolute rule: **it never lies**. Every sentence is generated from the record, and touching a sentence shows the check-up, plan or forecast it comes from. If the stream has no data, it says so. It never invents a cause, never names anyone, never gives medical or bathing advice, and never says the water is safe. That discipline is what turns the voice from a gimmick into an instrument scientists can respect. In a classroom it turns into a lesson: why does the stream sound sad? Who will go and look?

The physical world

Half of StreamChart's strength lies off the screen. The app generates a printable **sign** for each stream, styled like the chart clipped to a hospital bed. It shows the stream's name and current state, the sentence *"This stream is a patient. Take its pulse in one minute."*, a QR code that opens the check-up directly, and the spot to stand on so that every photo is taken from the same

viewpoint, which is what makes before/after comparisons possible. There is a **one-page school kit**: choose a stream near the school, four roles for the pupils (the photographer, the smell detective, the keeper of the file, the spokesperson), a 45-minute plan, and three questions to discuss back in class. Nothing to buy, nothing to install. Once a season, a city can hold a **check-up day** that brings every grey “not followed” stream back onto the map.

Built on standards: the chart is an HL7 FHIR R4 record (Track 7)

The medical metaphor isn't only a story; it is also a data model. Every stream's chart exports as an **HL7 FHIR R4 transaction Bundle**, and one button posts it to the public **HAPI FHIR R4 server**, which accepts it: 46 resources, 43 created and the rest matched, and the export is idempotent when re-posted.

- The stream is a **Patient** (carrying an extension that flags it as a non-human environmental subject) *and* a **Location** (its geography). This is the metaphor made literal, and a deliberate modelling decision. In R4, CarePlan, DiagnosticReport and RiskAssessment require a Patient or Group subject. Representing the reach as a Patient means any FHIR server, viewer or analytics pipeline can handle a river with no custom resource type.
- The 60-second form is a **Questionnaire**. Each check-up is a **QuestionnaireResponse** (carrying its seal hash), a **Media** for the photo, and one **Observation** per indicator with a confidence-grade extension, *derivedFrom* the questionnaire response and the photo, grouped in a panel Observation.
- The AI's look is a separate **Observation** produced by a **Device** (the model), carrying its own seal hash and a note saying whether it came from a live call or a recorded answer.
- A **Provenance** records who did what: the citizen as author, the AI as informant, with a signature carrying the seal.
- Each hypothesis is a **DiagnosticReport**: *preliminary* until signed, *final* afterwards.
- The seven-day indication is a **RiskAssessment** with a qualitative risk, a time window, its rationale and its method.
- The care plan is a **CarePlan** whose activities are coded with the D2.4 section numbers. It moves from *draft/proposal* to *active/plan* to *completed*, and removed lines become *cancelled* with the ecologist's reason. The co-signature is a **Provenance** with an ISO *Verification Signature* by the ecologist (**Practitioner**).
- Every reply to a citizen is a **Communication**.

Demonstration data is tagged as such in `meta.tag`. For researchers, every fact carries its provenance, including the fact that an AI took part and what it answered. That is what makes these observations reusable under FAIR principles.

Who it is for

- **Citizens and schools** who observe, and finally hear back.
- **Ecologists** who validate, and get a queue of cases sorted by difficulty, with photos, the AI's view, the citizens' view and a confidence level, so they can decide in ten seconds and sign their name to it.
- **Municipal services** that prioritise and act, with a ward room that tells them where to go first.
- **Public-health actors** who need early, cautious, explained signals rather than sirens.

- **Researchers** who need reusable data with provenance.
- **The OneAquaHealth consortium itself**, whose tools gain a common front door and a reason to be used after the project ends.

How I built it

StreamChart is a static single-page application in **React 18 and TypeScript**, built with **Vite**. It has no server and stores no secrets. The browser calls Google's Gemini API and Open-Meteo directly, so the app can be hosted anywhere for free. That is part of the feasibility argument: a city can fork it and run it tomorrow.

The core is a **pure TypeScript domain layer** with no UI code, which models the medical logic of a stream:

- `fields.ts` defines the check-up: the seven OneAquaHealth indicators, their options, which ones are abnormal, which ones a photo can judge, and the wording in three registers.
- `engine.ts` holds the clinical reasoning. It consolidates check-ups from the last three weeks into *findings* and grades each one A–D with explicit rules: photo, person–AI agreement, corroboration by distinct people, the reliability of the person's eye over past check-ups, expert validation, and contradictions within a week. It then computes the state (not followed, stable, under watch, alert), the hypotheses with their confidence and reasons, the seven-day indication, the three-line ward-room summary, the pulse series, and the stream's voice, sentence by sentence with sources.
- `world.ts` contains the actions that change the record (submit a check-up, sign a plan, record an intervention, update the forecast) and the side effects that close the loop: state-change events, hypotheses, the proposed plan, notifications, remission.
- `ai.ts` calls **Gemini vision** (`gemini-3.5-flash-lite` by default, selectable in Settings) with a **structured-output JSON schema**. The schema forces one of the allowed values or unsure for each field, a confidence, a short note, and up to four observations with bounding boxes normalised to 0–1000, so they can be drawn on the photo. The prompt forbids diagnosing, naming culprits or saying the water is safe or unsafe. The response is validated, clamped, and sealed with SHA-256 over canonical JSON before anyone can see it.
- `sha256.ts` is a small synchronous SHA-256 with canonical JSON serialisation, so seals are reproducible and the domain stays pure and testable.
- `fhir.ts` builds the FHIR R4 transaction Bundle described above, with deterministic UUIDs and conditional creates.
- `catalogue.ts` quotes the measures from the D2.4 catalogue with their section numbers.

Because the domain is pure, it is covered by tests. One of them tells the brief's whole story as an automated test: a forgotten stream, Mathis's first check-up (under watch, foam at grade B, odour at grade C), Léa's concordant check-up (odour rises to B, hypothesis appears), Inès's report of stagnant water, a heatwave forecast (seven-day indication, state goes to alert, plan proposed), the ecologist's signature (findings reach grade A, every contributor notified, the voice mentions her), an intervention, and a follow-up three weeks later (stable, remission, thank-you message to the first reporter). The same test checks that the FHIR bundle contains every expected resource type.

AI: bring your own key. The Settings page accepts a Google AI Studio key, stored only in the user's browser and sent only to Google. Judges shouldn't need a key to see the product work, so for the eight bundled sample photos I recorded **real Gemini answers** in advance with a script that uses exactly the same prompt and schema as the app. Without a key, those recorded answers are replayed and **labelled as recorded** wherever they appear. For a user's own photo without a key, a simple offline colour analysis gives a minimal answer, again clearly labelled and never presented as AI vision.

Photos. I didn't want stock imagery or generated pictures pretending to be real. Every photo in the app is a **real urban stream from Wikimedia Commons** under a Creative Commons licence, most of them from Toulouse (the Marcaissonne, the Sausse, the Maltemps brook), credited in the app and the README. The demonstration streams use them as stand-ins, and the app says that the states shown make no claim about those places.

Design. The interface follows a calm, airy dashboard design system: a cool blue-grey canvas, a framed app, white glass-like cards, Inter at small sizes, a blue accent, soft status pills with icons (stable, under watch, alert, not followed), tapered confidence bars, and a single warm gradient for the most important gesture on each screen. There is a light and a dark theme. The check-up is phone-first, with very large touch targets.

The emergency scene. To show the whole loop within the attention span of a judge, the app includes a 60-second accelerated simulation of one week in the life of a stream. It is labelled *"SIMULATION — accelerated time"* from the first frame. The people and the heatwave are scripted, but every grade, state, hypothesis, risk, plan, signature and notification on screen is computed live by the real engine. At the end, the simulated record can be opened and explored like any other chart.

Tools. I used Playwright to drive the app end to end for visual QA and for the screenshots, ffmpeg to assemble the demo video, and Fish Audio for the narration. Claude Code was my pair-programmer throughout.

Challenges I ran into

Keeping the metaphor honest. The medical metaphor is powerful, but it could easily slide into pretending: a "diagnosis" from a photo, a "validated" label nobody earned, an alarming red screen. I wrote four rules (hypothesis not diagnosis, a human signs, no drama, no blame) and enforced them in code, not just in copy. The system literally can't write *validated* without a signature, the voice can only speak from the record, and the health indication always carries its reassessment date and its disclaimer.

Designing grades that move for the right reasons. My first version treated any different answer as a contradiction, so a stream that was clear three weeks ago made today's foam look doubtful. Real streams change. I had to separate *contradiction* (a different answer within a week) from *history* (an earlier state). Odour needed its own rule, since a photo can't corroborate a smell, so a single nose stays at grade C until someone else confirms it. Expert validation had to be scoped to the fields the ecologist actually confirmed, not to everything in the same check-up.

Making the AI humble. A vision model asked to fill a form will happily fill every field. The structured-output schema always offers an unsure value, the prompt treats “unsure” as good practice, and odour is forced to unsure. Seeing the model answer “unsure” for water clarity when the surface was covered in foam was one of the moments where the design felt right.

Making the comparison trustworthy. If the citizen sees the AI's answer first, the comparison means nothing. Sealing both answers with hashes before the reveal costs almost nothing to implement, and it turns a UX intention into a verifiable property. The seals are shown in the expert details and exported in the FHIR Provenance.

Mapping a river onto FHIR. FHIR R4 has no resource for an ecosystem. Modelling the stream only as a Location would have been tidy, but CarePlan, DiagnosticReport and RiskAssessment need a Patient or Group subject. Representing the reach as a Patient, flagged by an extension and paired with a Location, keeps every standard workflow resource usable. I also had to make the export idempotent: the public HAPI server rejects exact duplicates, so stable resources use conditional creates on their identifiers.

Being honest about what is real. A demo works better when every part of it is real, but some parts can't be real in a prototype: a real ecologist's signature, weeks of follow-up, five cities competing. I chose to label every simulated element where it appears: a *demonstration data* pill in the header, a *simulation* banner on the emergency scene, *illustrative* tags on the Stream Cup, *role-play* next to the ecologist's name. There is also an *Honesty & sources* page listing what is real, what is demonstration data and what we do not claim.

Accomplishments I'm proud of

- **The sealed second look.** Citizen first, AI sealed, comparison second, and the last word human, with verifiable hashes. It addresses Track 3's worry about AI replacing human judgement and Track 1's worry about inconsistent observations in a single gesture.
- **A co-signature you can actually see.** The human in the loop isn't a sentence in a slide. It is a button, a stamp, a name in the record and a Provenance resource.
- **A loop that closes.** Almost every citizen-science tool handles the observation; StreamChart handles the reply, all the way to remission and a thank-you to the first person who noticed.
- **A voice that never lies.** A first-person stream that moves children and still holds up to scientific scrutiny, because every sentence links back to its source.
- **Real standards, really accepted.** A FHIR R4 export that a public FHIR server accepts, with a data model that makes every fact traceable.
- **Respect for the project's own work.** The indicators are OneAquaHealth's, the measures are quoted from their D2.4 catalogue with section numbers, and DipteraCAST and the DSS have a natural place in the chart. StreamChart is meant as a common front door for the tools the consortium already built, not a competitor to them.

What I learned

The most important feature of a citizen-science tool is the reply. A grade, a hypothesis and a dashboard are all means to one end: telling the person who looked what their looking changed. Every design decision got easier once I asked *what does this change for the patient, and who gets told?*

Humility can be designed. "Unsure" answers, confidence grades, a "why?" button, a reassessment date, and a disagreement shown rather than hidden all cost very little to build, and together they make a system that scientists can trust and citizens aren't intimidated by.

Standards can carry a story. Expressing a stream's chart in FHIR wasn't a compliance exercise. It made the metaphor rigorous: if the stream is a patient, then the plan is a CarePlan, the signature is a Provenance and the reply is a Communication. The narrative and the data model turned out to be the same thing.

Finally, the physical world matters. A sign at the foot of a stream, the same photo viewpoint every time, and a one-page school kit will probably do more for participation than any feature I could add to the app.

What's next for StreamChart

- **A pilot** with one school and one municipal service in one of the five research cities. Toulouse is the natural first, with signs installed on two or three real streams.
- **A validation study of the confidence grades** against expert assessments: do A-grade findings hold up, and how should the rules be tuned?
- **Plugging into the OneAquaHealth tools:** DipteraCAST predictions as the chart's laboratory analysis, the Decision Support System as the source of care plans, and the Citizen Science App's data model as the check-up's back end. StreamChart is designed to plug into these tools; it isn't connected to them yet.
- **Full multilingual support** for the five cities (French, Portuguese, Dutch, Norwegian, Italian), plus English, using the same three-register structure.
- **The Stream Cup:** a seasonal challenge between the five cities that measures what matters (share of streams followed, remissions, average grade, school participation) rather than raw volume. It is sketched in the app and labelled as roadmap.
- **An open template** so that any city adopting the same indicators can publish its streams as patients.

Honesty: what is real, what is demonstration

Real and reproducible: the 60-second check-up with your own photo or a real sample photo; the sealed comparison with the AI (live Gemini vision with your key, real recorded Gemini answers without one) and its SHA-256 seals; the confidence grades computed by displayed rules; states, hypotheses and the care-plan proposal computed by the engine; the seven-day indication reading a live Open-Meteo forecast; the co-signature and the change from *proposed* to *validated*; the replies to citizens; the stream's voice, grounded sentence by sentence; and the HL7 FHIR R4 export accepted by the public HAPI FHIR server.

Demonstration data, labelled as such: the people (Mathis, Léa, Inès, Camille and others) and their past check-ups; the referent ecologist, Dr Ferreira, who is a role-play (no real ecologist has signed anything); the scripted heatwave and citizens of the emergency scene, which runs in accelerated time on the real engine; the Stream Cup figures for the four other cities; and the background measures shown in timelines. The photos are real urban streams used as stand-ins, and the states shown make no claim about those places.

Not claimed: any integration with OneAquaHealth systems beyond public information and compatibility with their indicators and catalogue; any diagnosis or epidemiological forecast; any validation of the grading method; any official relationship with a city or a school.

If you only remember one sentence, I hope it is the one I'd love to hear judges repeat to each other over coffee: *they gave a medical record to a river.*

Pitch deck



OneAquaHealth IEEE Global Hackathon 2026 - Track 7 Digital Health Standards

StreamChart

Every patient has a chart.
Now every stream does.

A health record and a care loop for every urban stream: a 60-second citizen check-up, a sealed AI second look, confidence grades, a care plan co-signed by an ecologist — and a reply to the citizen. Exported as HL7 FHIR R4.



streamchart-oneaquahealth.netlify.app

github.com/chessoren/streamchart

- The problem

Citizen science gives a lot — and never hears back.

01

No continuous thread

Observations pile up without a story. Nobody can say “this stream was better in May, relapsed in August, here is why”.

02

No measured trust

A child's photo and an ecologist's survey land in the same database with the same apparent weight. Authorities can't tell how far to trust either.

03

No reply

The person who gave their time receives nothing, so they don't come back. Track 5 names it: **low repeat engagement**.

OneAquaHealth has built the organs of a health system for urban streams — citizen app, city dashboards, resilience map, GEOSSIP, a decision-support system, DipteraCAST. **What is missing is the patient.**

StreamChart

2 / 13

Slide 2 of 13

- The insight

Medicine solved this a century ago.

Three inventions: the **medical record** that keeps the story, the **level of evidence** that says how far to trust each fact, and the **care loop** that links a symptom to an action and a follow-up.

From streams
to patients
to systems.

Patient	A stream reach
Medical record	The stream's chart
Consultation	A 60-second check-up
Level of evidence	Confidence grade A–D
Diagnosis	Hypothesis — never certainty
Prescription	Care plan, co-signed
Follow-up visit	Same-viewpoint check-up
Recovery	Remission

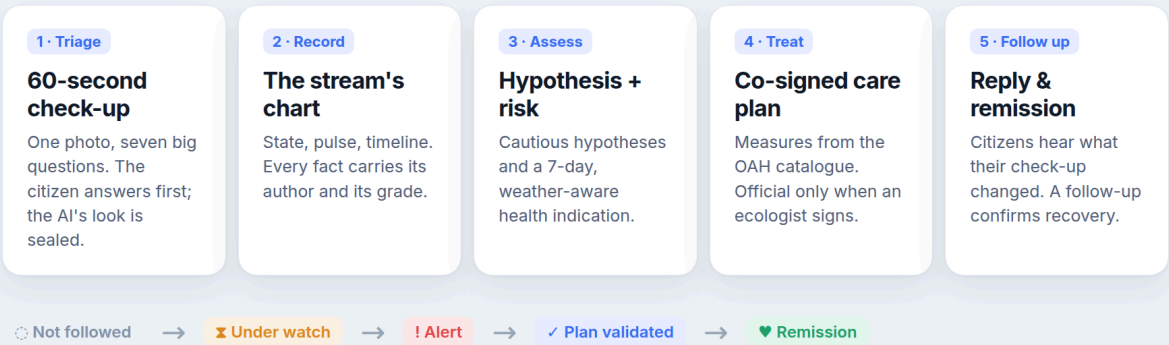
StreamChart

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Slide 3 of 13

• How it works

One loop: observe, understand, act, come back.

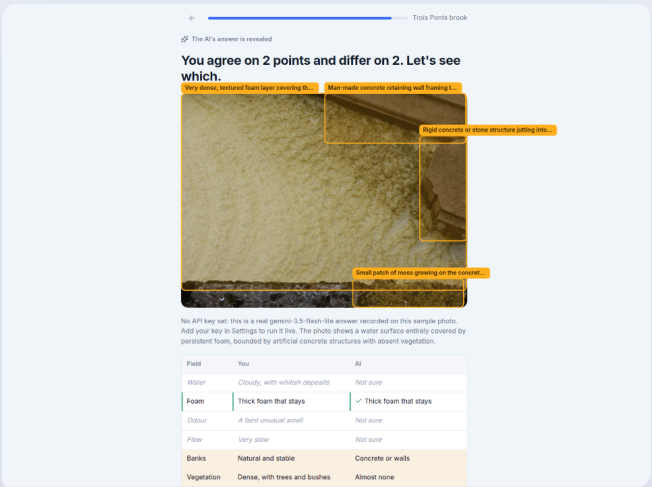


Slide 4 of 13

• Track 3 · AI that assists, never replaces

The citizen answers first. The AI's answer is sealed.

- Gemini vision looks at the photo while the citizen answers — it never sees their answers.
- Its answer is hashed (SHA-256) on arrival and revealed only after the citizen seals their own: **commit–reveal**.
- Differences are highlighted on the photo. Keep mine, adopt the AI's, or ask a second opinion. **The last word is human.**
- The AI may say “unsure” — and always does for odour.



Slide 5 of 13

• Measured trust

Every fact carries a confidence grade.

- A** Photo, person and AI agree, corroborated by others — or validated by an expert
- B** Photo with person–AI agreement, or two concordant people
- C** Plausible but isolated, or an unresolved disagreement
- D** Fragile: no photo, imprecise or contradicted

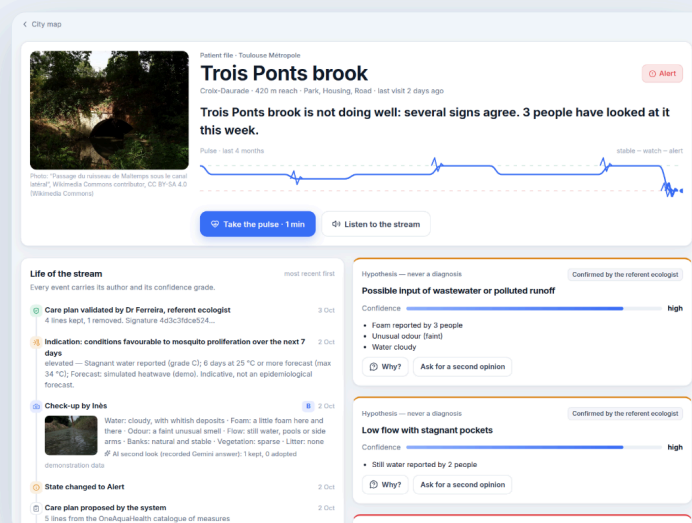
The grade is a property of the information, not a judgement of the person. It moves: a **C** becomes **B** when a neighbour confirms it, and **A** when the referent ecologist validates it.

The community becomes more credible than the sum of its parts.

Disagreements are shown, never smoothed. A proposed method — validation against expert assessments is on the roadmap.

Slide 6 of 13

• Track 2 - Data to insight



A stream's whole life, on one page.

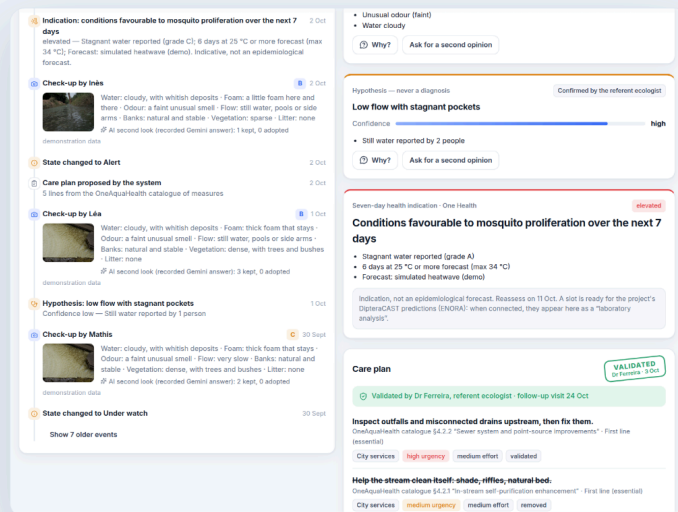
- Identity, state badge and one sentence anyone can read.
- A pulse: a heartbeat trace of its health over four months.
- A timeline of check-ups, measures, hypotheses, plans and interventions — with authors and grades.
- Readable by a ten-year-old, credible to an ecologist. Expert details one tap away.

Slide 7 of 13

• Track 6 · The One Health bridge

From a stream's symptoms to a signal for people.

- **Hypotheses, never diagnoses:** “possible input of wastewater or polluted runoff — confidence medium”, with reasons and what would confirm it.
- **Seven-day indication:** stagnant water × a live **Open-Meteo** forecast → conditions favourable to mosquitoes. Explained, dated, labelled “not an epidemiological forecast”.
- A slot is ready for **DipteraCAST** (ENORA) as the chart's “laboratory analysis”.

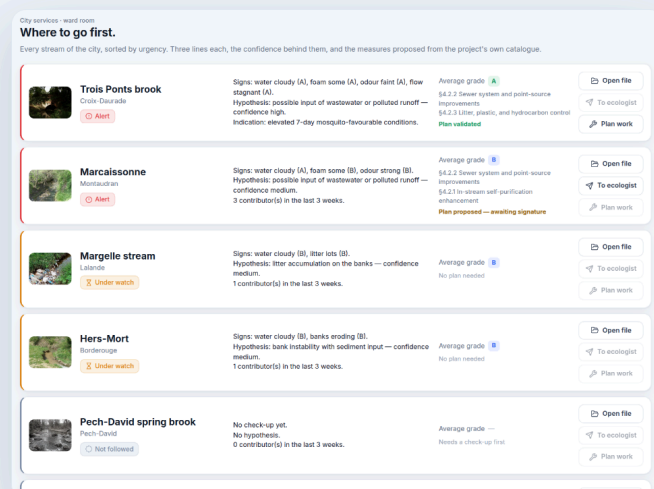


StreamChart

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• Human in the loop, made visible



StreamChart

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Slide 9 of 13

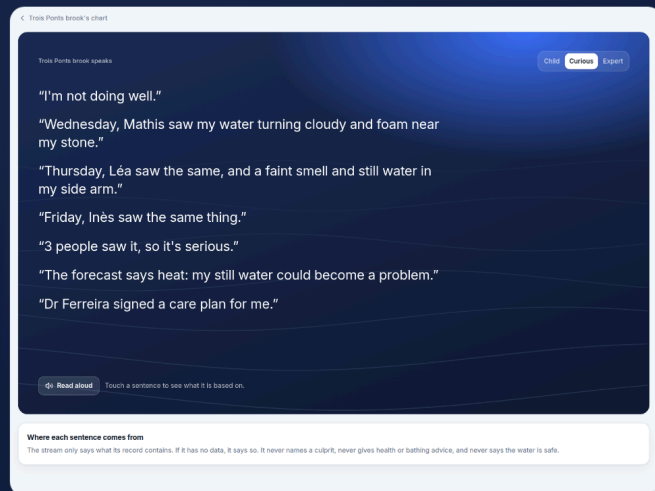
The AI proposes. An ecologist signs.

- Care plans of at most five lines, quoted from the **OneAquaHealth D2.4 Catalogue of measures** (§4.2.2 Sewer system and point-source improvements...).
- Each line: what, why, who (city, association, residents), urgency, effort.
- The plan stays *proposed* until the referent ecologist removes what is premature and **signs** — the stamp goes from proposed to validated.
- The city's **ward room** shows where to go first, in three lines.

"Your check-up triggered an inspection."

Every contributor hears what their check-up changed: compared, first in the chain, plan validated, intervention done, follow-up, remission. No points, no streaks — **the only reward that lasts is seeing your gesture have an effect.**

And the stream speaks, in the first person, in three registers — grounded sentence by sentence in its record. Tap a sentence, see its source. It never accuses, never alarms, never says the water is safe.



The chart is an HL7 FHIR R4 record.

The stream	Patient + Location
60-second form	Questionnaire / QuestionnaireResponse
Each answer · photo	Observation (+grade) · Media
The AI's look	Observation from a Device
Who did what · seals	Provenance + signature
Hypothesis	DiagnosticReport
7-day indication	RiskAssessment
Care plan · co-signature	CarePlan · Provenance (verifier)
Reply to the citizen	Communication

One click exports a transaction Bundle — and posts it to the public **HAPI FHIR R4** server, which accepts it (43 resources created, re-posts idempotent).

Making the stream a **Patient** is the metaphor made literal: CarePlan, DiagnosticReport and RiskAssessment need a Patient subject, so any FHIR server, viewer or pipeline handles a river with no custom resource.

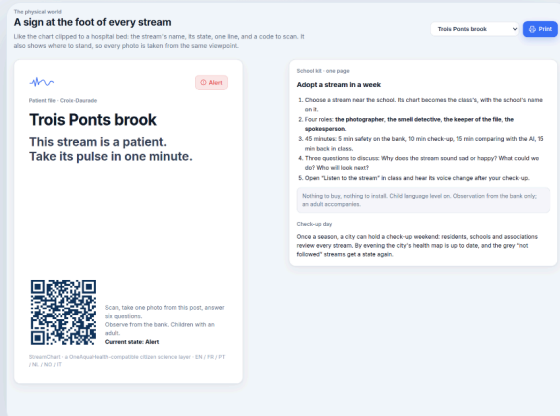
FAIR-ready provenance

Accepted by HAPI FHIR

Demo data tagged in meta

• Feasibility & scale

No hardware. No budget. Five cities tomorrow.



A sign per stream

A printed "patient file" with a QR code — and the same viewpoint for every photo.

A school kit

One page, 45 minutes, four roles. Any school adopts a stream in a week.

Built on OAH

Their indicators, their catalogue, their tools as "labs". A layer, not a competitor.

Static & open

React + TypeScript, no server, BYO Gemini key, MIT. Any city can fork it.

Benevento · Coimbra · Ghent · Oslo · Toulouse — a seasonal **Stream Cup** on what matters: streams followed, remissions, quality, schools.

StreamChart

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Slide 12 of 13

• Honest by design · What's next

Real where it matters. Labelled where it isn't.

Real: check-up, sealed Gemini comparison, grades, states, hypotheses, live weather, co-signature, replies, voice, FHIR export.

Demonstration: the people and past check-ups, the role-played ecologist, the scripted heatwave of the emergency scene. Photos are real Toulouse-area streams (CC BY-SA).

Next: a pilot with one school and one city service; a validation study of the grades; plugging in DipteraCAST and the DSS.



"They gave a medical record to a river."

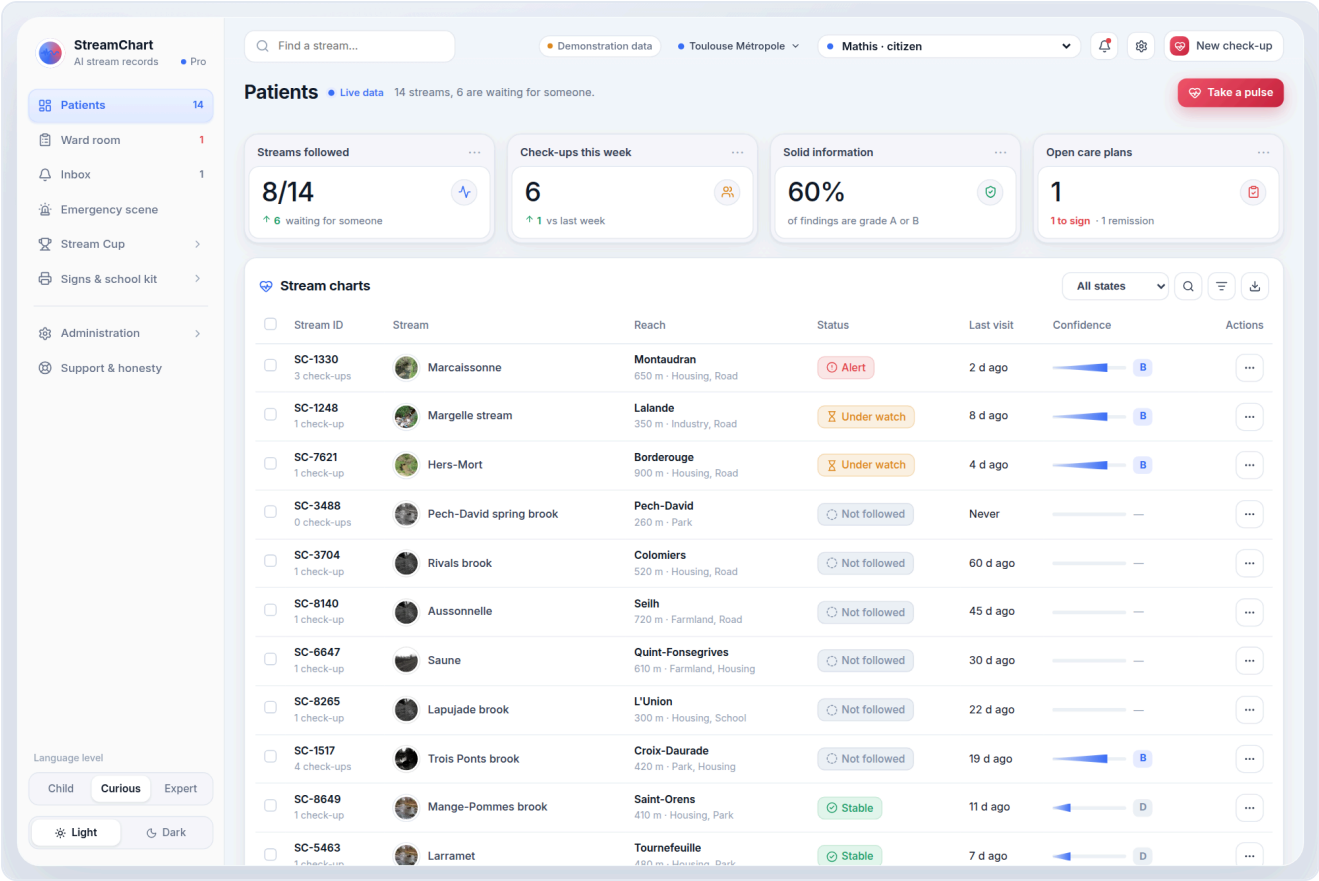
streamchart-oneaquahealth.netlify.app

github.com/chessoren/streamchart

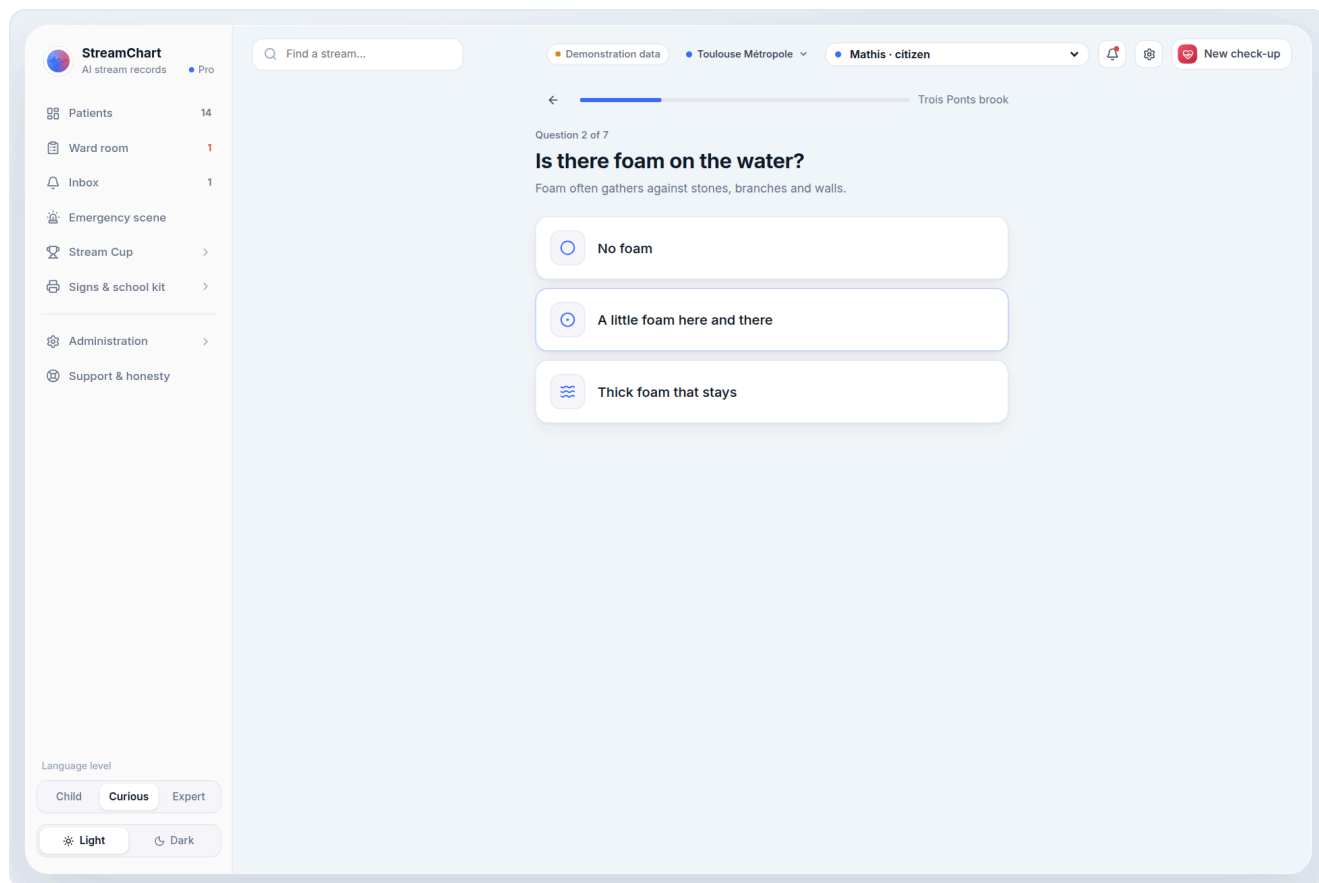
13 / 13

Slide 13 of 13

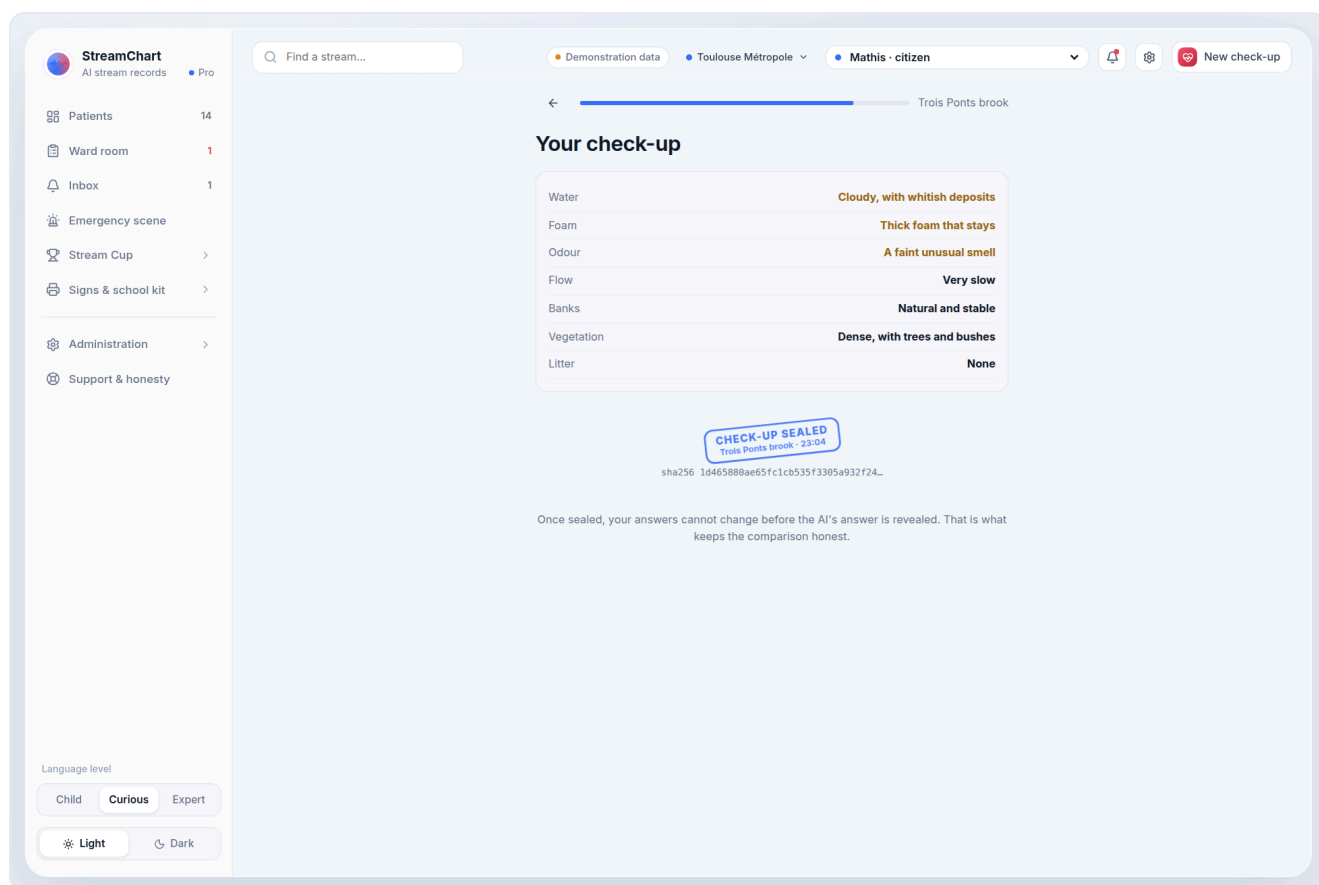
Screenshots



Patients dashboard



Checkup question



Checkup sealed

StreamChart

AI stream records

Pro

Patients

14

Ward room

1

Inbox

1

Emergency scene

Stream Cup

Signs & school kit

Administration

Support & honesty

Find a stream...

Demonstration data

Toulouse Métropole

Mathis - citizen

New check-up

Trois Pons brook

The AI's answer is revealed

You agree on 2 points and differ on 2. Let's see which.

Very dense, textured foam layer covering th...

Man-made concrete retaining wall framing L...

Rigid concrete or stone structure jutting into...

Small patch of moss growing on the concret...

No API key set: this is a real gemini-3.5-flash-lite answer recorded on this sample photo. Add your key in Settings to run it live. The photo shows a water surface entirely covered by persistent foam, bounded by artificial concrete structures with absent vegetation.

Field	You	AI
Water	Cloudy, with whitish deposits	Not sure
Foam	Thick foam that stays	✓ Thick foam that stays
Odour	A faint unusual smell	Not sure
Flow	Very slow	Not sure
Banks	Natural and stable	Concrete or walls
Vegetation	Dense, with trees and bushes	Almost none

Ai reveal

StreamChart

AI stream records

Pro

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Ward room

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Inbox

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Emergency scene

Stream Cup

Signs & school kit

Administration

Support & honesty

Find a stream...

Demonstration data

Toulouse Métropole

Mathis - citizen

New check-up

Here: Man-made concrete retaining wall framing the water body.
Here: Small patch of moss growing on the concrete edge.

Banks: you said "Natural and stable", the AI says "Concrete or walls" — Concrete or man-made structures visible along the top and bottom edges.

✓ I keep my answer

I adopt the AI's

Ask for a second opinion

Vegetation: you said "Dense, with trees and bushes", the AI says "Almost none" — No aquatic or bank vegetation is visible except minor moss on concrete structures.

✓ I keep my answer

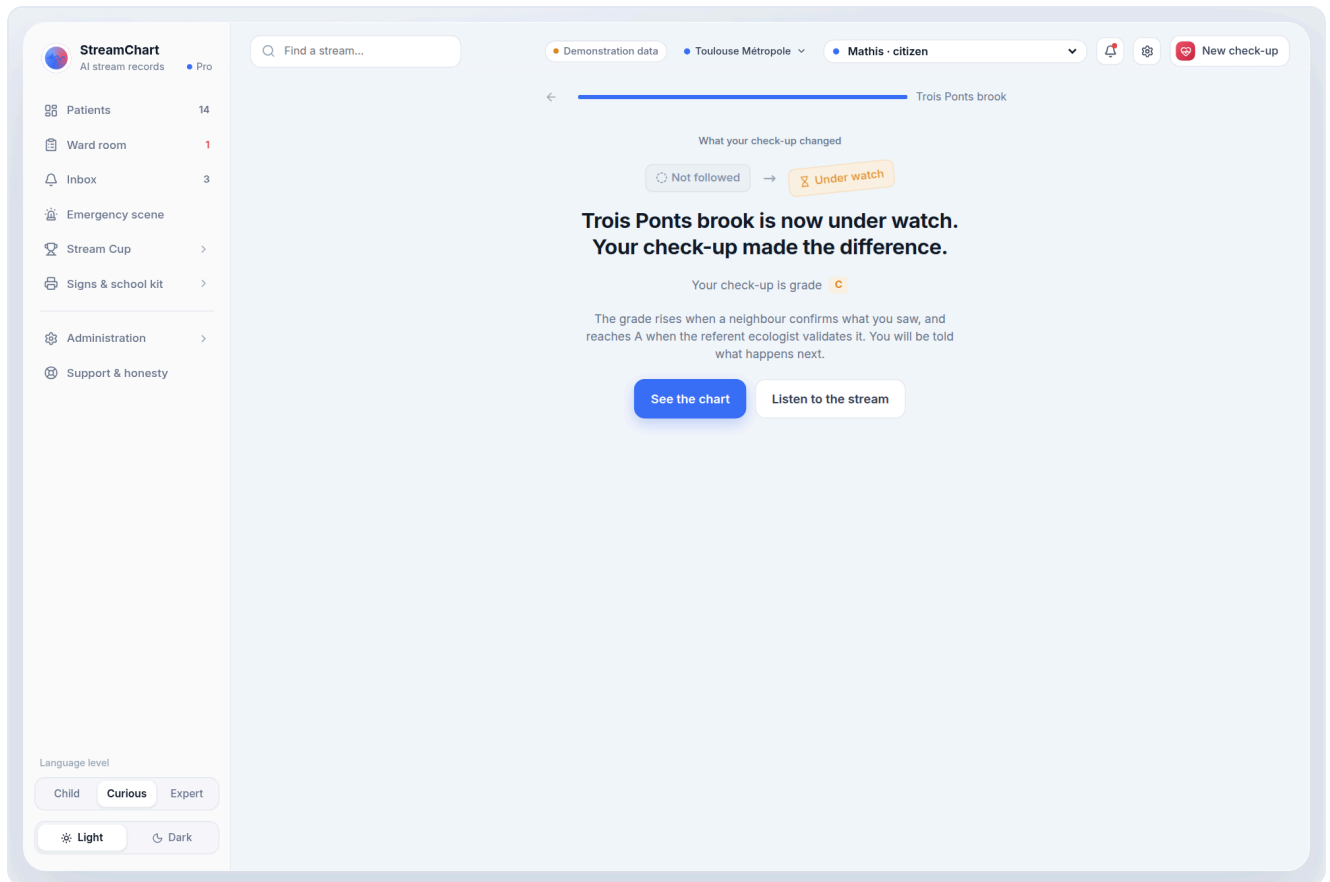
I adopt the AI's

Ask for a second opinion

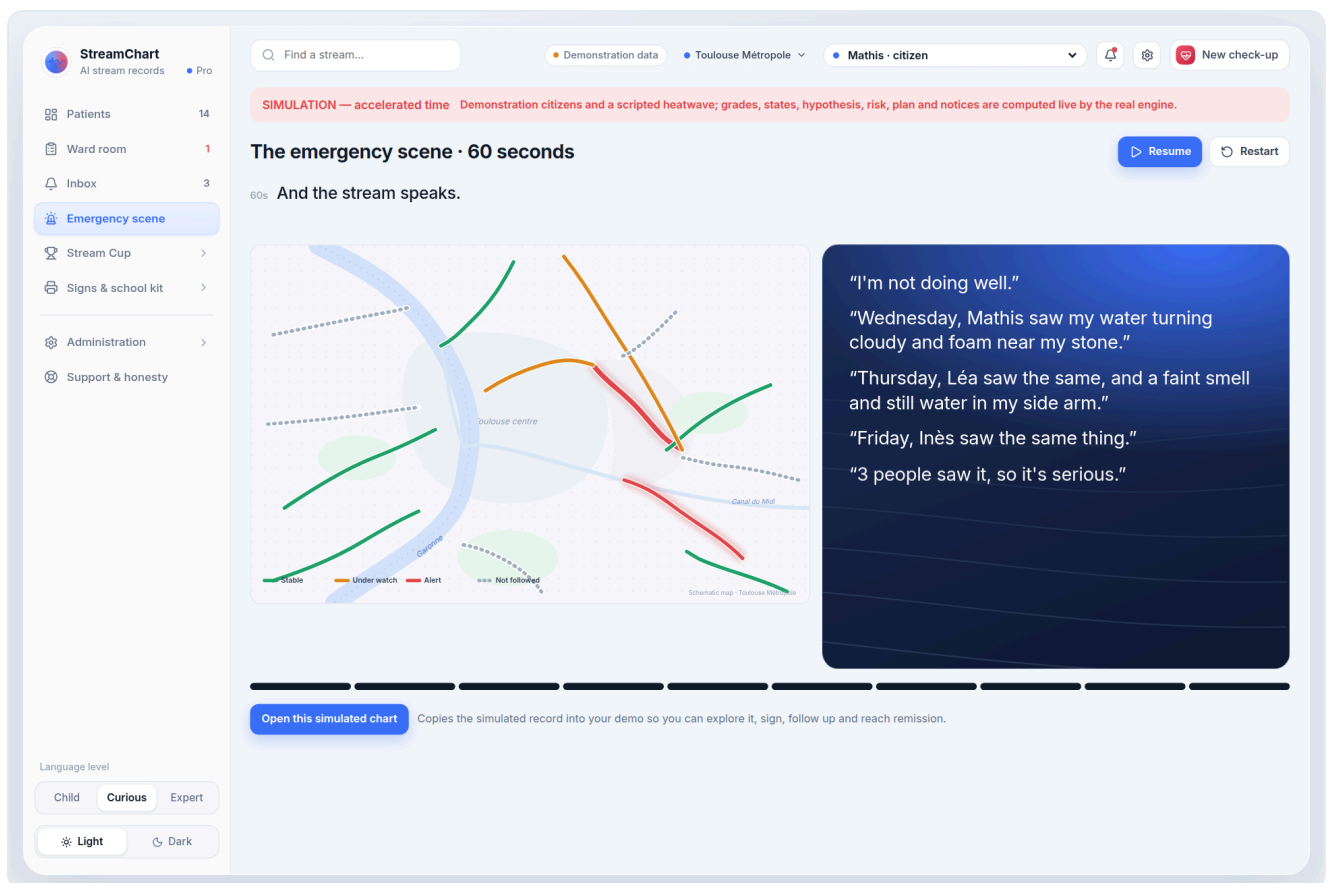
Your seal sha256 1d465880ae5fc1cb535... · AI seal sha256 6b29ce892766c0d7420a... (computed when the AI answered, before the reveal)

Add my check-up to the chart

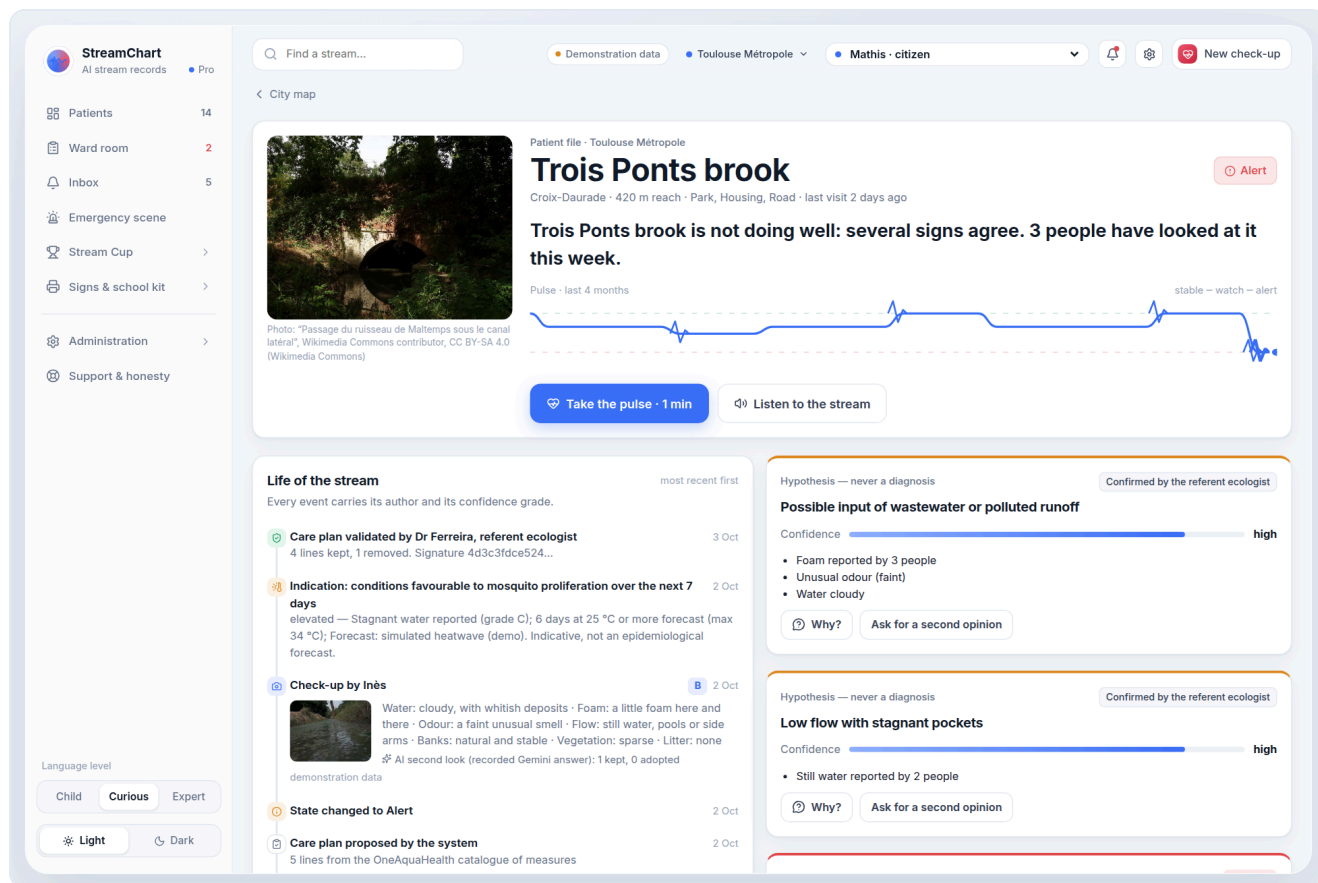
Ai compare decide



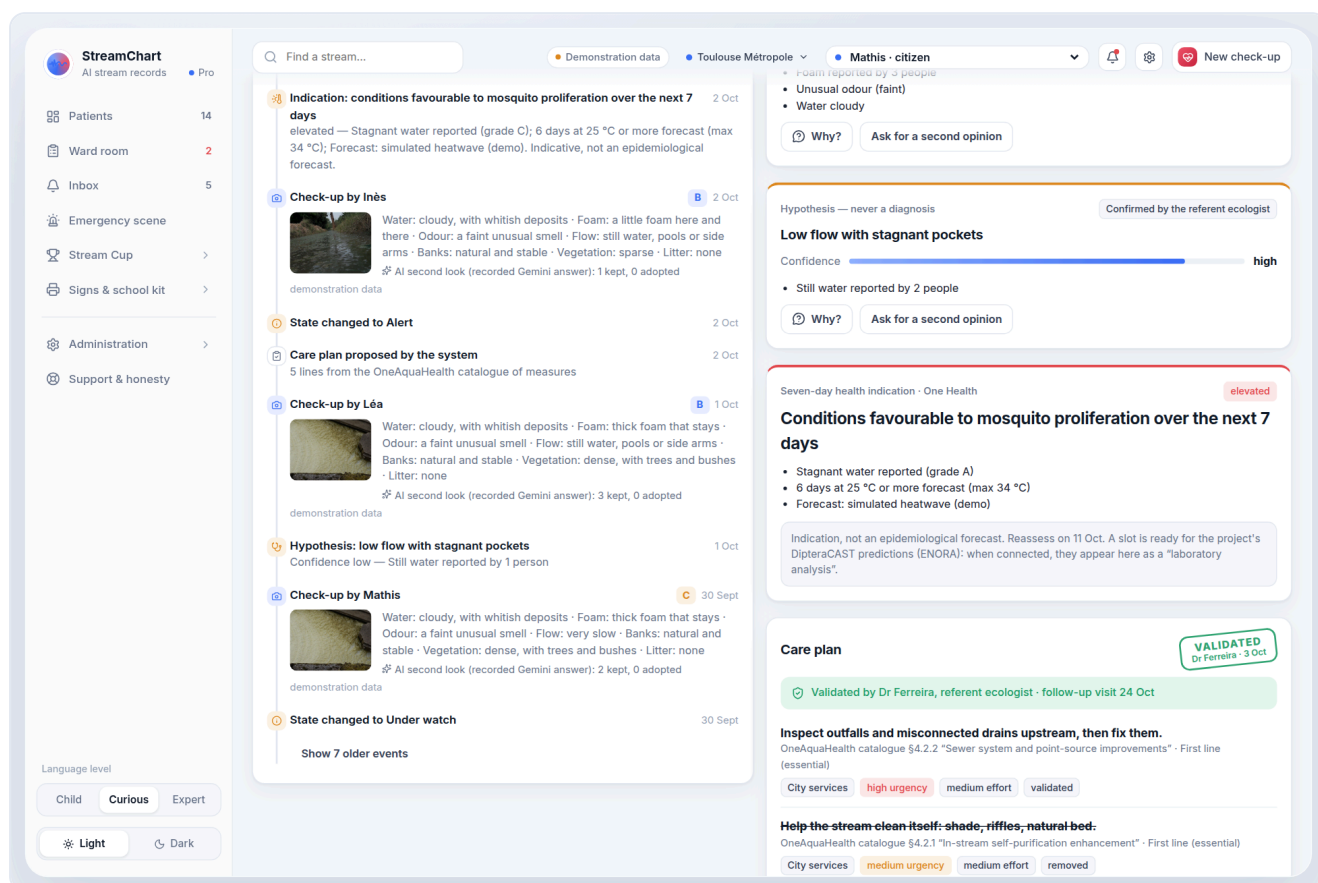
What your checkup changed



Emergency scene voice



Stream chart alert



Hypothesis risk careplan

StreamChart

AI stream records

Pro

Patients

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Ward room

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Inbox

5

Emergency scene

Stream Cup

Signs & school kit

Administration

Support & honesty

Find a stream...

Demonstration data

Toulouse Métropole

Mathis - citizen

New check-up

City services - ward room

Where to go first.

Every stream of the city, sorted by urgency. Three lines each, the confidence behind them, and the measures proposed from the project's own catalogue.

Trois Pons brook

Croix-Daurade

Alert

Signs: water cloudy (A), foam some (A), odour faint (A), flow stagnant (A).

Hypothesis: possible input of wastewater or polluted runoff — confidence high.

Indication: elevated 7-day mosquito-favourable conditions.

Average grade

A

\$4.2.2 Sewer system and point-source improvements

\$4.2.3 Litter, plastic, and hydrocarbon control

Plan validated

Open file

To ecologist

Plan work

Marcaissonne

Montaudran

Alert

Signs: water cloudy (A), foam some (B), odour strong (B).

Hypothesis: possible input of wastewater or polluted runoff — confidence medium.

3 contributor(s) in the last 3 weeks.

Average grade

B

\$4.2.2 Sewer system and point-source improvements

\$4.2.1 In-stream self-purification enhancement

Plan proposed — awaiting signature

Open file

To ecologist

Plan work

Margelle stream

Lalande

Under watch

Signs: water cloudy (B), litter lots (B).

Hypothesis: litter accumulation on the banks — confidence medium.

1 contributor(s) in the last 3 weeks.

Average grade

B

No plan needed

Open file

To ecologist

Plan work

Hers-Mort

Borderouge

Under watch

Signs: water cloudy (B), banks eroding (B).

Hypothesis: bank instability with sediment input — confidence medium.

1 contributor(s) in the last 3 weeks.

Average grade

B

No plan needed

Open file

To ecologist

Plan work

Pech-David spring brook

Pech-David

Not followed

No check-up yet.

No hypothesis.

0 contributor(s) in the last 3 weeks.

Average grade

—

Needs a check-up first

Open file

To ecologist

Plan work

Language level

Child

Curious

Expert

Light

Dark

Ward room

StreamChart

AI stream records

Pro

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Ward room

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Emergency scene

Stream Cup

Signs & school kit

Administration

Support & honesty

Find a stream...

Demonstration data

Toulouse Métropole

Mathis - citizen

New check-up

Mathis's inbox

What your check-ups changed.

No points, no streaks. Just what happened because you looked.

Sausse

Welcome. The streams near you are listed on the map. Grey ones are waiting for someone.

demonstration data

4 Oct, 23:04

Trois Pons brook

Your check-up on Wednesday triggered an inspection. Dr Ferreira validated the care plan for Trois Pons brook. A follow-up visit is planned in three weeks. You are part of its care team.

demonstration data

3 Oct, 23:04

Trois Pons brook

Trois Pons brook is now alert.

2 Oct, 05:04

Trois Pons brook

Your check-up was compared with 1 other from the last three weeks. It is grade C.

demonstration data

30 Sept, 23:04

Trois Pons brook

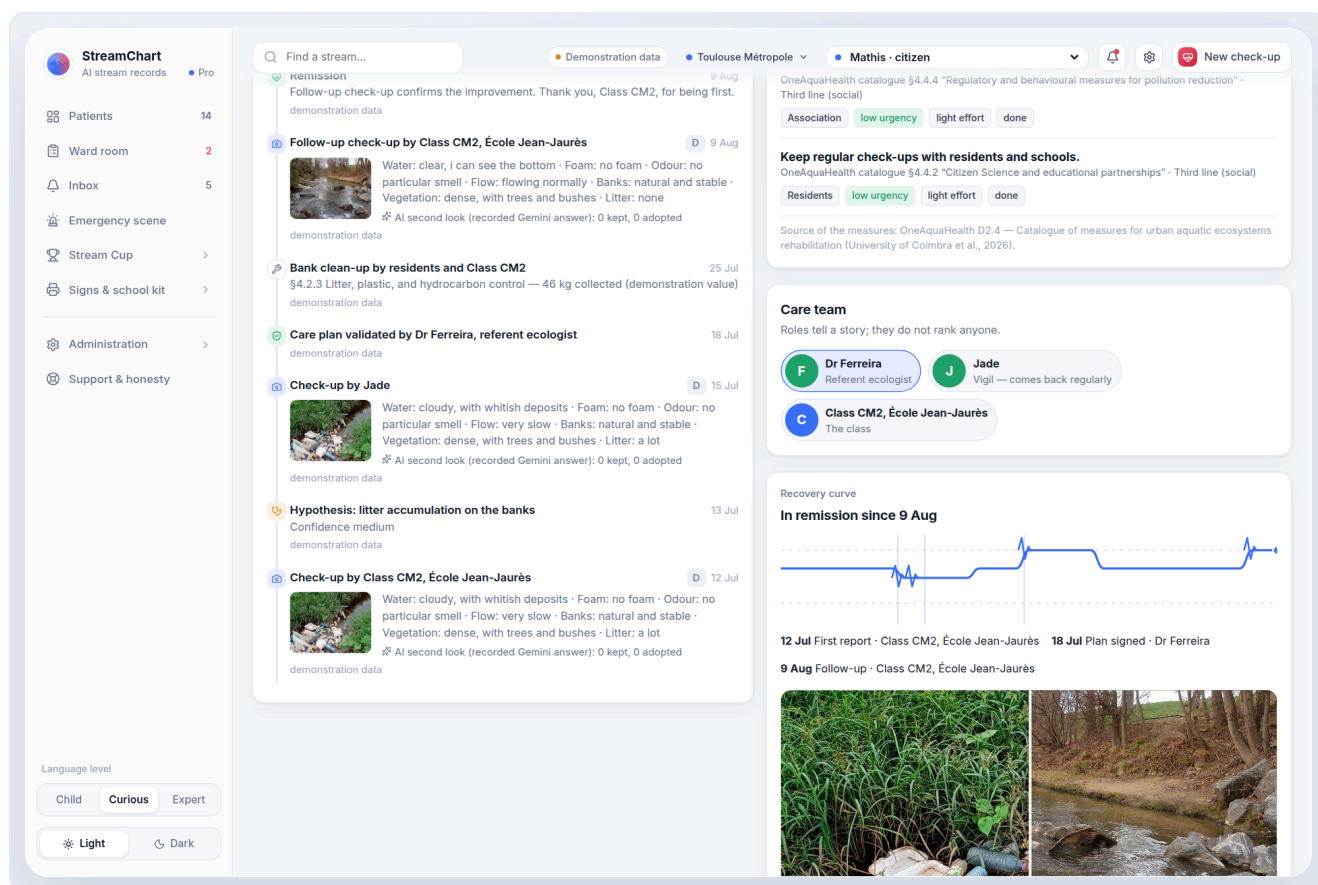
Trois Pons brook is now under watch. Your report is the first in the chain.

30 Sept, 23:04

Citizen inbox



Voice of the stream



Remission recovery

StreamChart

AI stream records

Pro

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Emergency scene

Stream Cup

>

Signs & school kit

>

Administration

>

Support & honesty

Find a stream...

Demonstration data

Toulouse Métropole

Mathis - citizen

Scout — eye close to consensus

New check-up

Details for experts

findings, grades, provenance, HL7 FHIR R4 export

Current findings (last 3 weeks)

Water	Elevated turbidity	photo available, person and AI agree, 3 people concordant, reliable eye (consistent past check-ups), validated by the referent ecologist
Foam	Localised, transient foam	photo available, person and AI agree, 3 people concordant, reliable eye (consistent past check-ups), validated by the referent ecologist
Odour	Faint (musty, earthy)	photo available, 3 people concordant, reliable eye (consistent past check-ups), validated by the referent ecologist
Flow	Stagnant pools / lentic patches	photo available, 2 people concordant, reliable eye (consistent past check-ups), validated by the referent ecologist, 1 contradicting check-up
Banks	Natural, stable banks	photo available, person and AI agree, 4 people concordant, reliable eye (consistent past check-ups)
Vegetation	Discontinuous cover	photo available, person and AI agree, 4 people concordant, reliable eye (consistent past check-ups)
Litter	None observed	photo available, person and AI agree, 4 people concordant, reliable eye (consistent past check-ups)

How grades are given

A

 Photo, person and AI agree, corroborated by others — or validated by an expert

B

 Photo with person-AI agreement, or two concordant people

C

 Plausible but isolated: one person, or an unresolved disagreement

D

 Fragile: no photo, imprecise or contradicted

Grades are a proposed method, not a validated one. A validation study against expert assessments is on the roadmap.

Seals (commit-reveal)

2 Oct, 05:04 · Inès · grade A

citizen sha256 65ce896cb6fc09257ff04a9a... · AI sha256 ce83052b30173f7ae18fa2f6...

1 Oct, 23:04 · Léa · grade A

citizen sha256 692272c632927790e550c15e... · AI sha256 6b29ce892766c0d7420a1855...

30 Sept, 23:04 · Mathis · grade A

citizen sha256 7c2eae6992a4c03bf862ca9e... · AI sha256 6b29ce892766c0d7420a1855...

15 Sept, 19:04 · Class CM2, École Jean-Jaurès · grade A

citizen sha256 f000cc6d1a0107aa9bbcaee... · AI sha256 897f1a60cb3ba4c31f7dd7e4...

HL7 FHIR R4 export

The chart as a transaction Bundle: the stream is a Patient and a Location; each check-up is a QuestionnaireResponse, Media and Observations with a confidence-grade extension; the AI's look is an Observation from a Device; hypotheses are DiagnosticReports, the 7-day indication a RiskAssessment, the care plan a CarePlan whose co-signature is a Provenance with a verification signature; replies to citizens are Communications.

Organization × 1

Location × 1

Patient × 1

Device × 1

Questionnaire × 1

Practitioner × 7

QuestionnaireResponse × 7

Media × 6

Observation × 62

Provenance × 7

DiagnosticReport × 2

RiskAssessment × 1

CarePlan × 1

Communication × 11

Download Bundle (.json)

Post to public HAPI FHIR server

Could not reach the public test server (Failed to fetch). The Bundle can still be downloaded.

Laboratory analyses

Designed to plug into the OneAquaHealth tools: DipteraCAST mosquito predictions and the Decision Support System. Not connected in this prototype; the seven-day indication uses a transparent rule with a live Open-Metoo forecast.

State: Alert. All people, check-ups and plans shown are demonstration data.

Language level

Child

Curious

Expert

Light

Dark

Fhir r4 export

StreamChart

AI stream records

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Emergency scene

Stream Cup

>

Signs & school kit

>

Administration

>

Support & honesty

Find a stream...

Demonstration data

Toulouse Métropole

Mathis - citizen

Trois Pons brook

Print

The physical world

A sign at the foot of every stream

Like the chart clipped to a hospital bed: the stream's name, its state, one line, and a code to scan. It also shows where to stand, so every photo is taken from the same viewpoint.

Patient file · Croix-Daurade

Alert

Trois Pons brook

This stream is a patient.

Take its pulse in one minute.

StreamChart · a OneAquaHealth-compatible citizen science layer · EN / FR / PT / NL / NO / IT

School kit · one page

Adopt a stream in a week

1. Choose a stream near the school. Its chart becomes the class's, with the school's name on it.

2. Four roles: the photographer, the smell detective, the keeper of the file, the spokesperson.

3. 45 minutes: 5 min safety on the bank, 10 min check-up, 15 min comparing with the AI, 15 min back in class.

4. Three questions to discuss: Why does the stream sound sad or happy? What could we do? Who will look next?

5. Open "Listen to the stream" in class and hear its voice change after your check-up.

Nothing to buy, nothing to install. Child language level on. Observation from the bank only; an adult accompanies.

Check-up day

Once a season, a city can hold a check-up weekend: residents, schools and associations review every stream. By evening the city's health map is up to date, and the grey "not followed" streams get a state again.

Sign and school kit

StreamChart

AI stream records

Pro

Patients

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Ward room

2

Inbox

5

Emergency scene

Stream Cup

Signs & school kit

Administration

Support & honesty

Find a stream...

Demonstration data

Toulouse Métropole

Mathis · citizen

New check-up

Honesty & sources

When in doubt between an impressive sentence and an exact one, we chose the exact one.

Real and reproducible

- The 60-second check-up, with your own photo or a real sample photo
- The sealed comparison with the AI (live Gemini vision with your key; real recorded Gemini answers without one) and its SHA-256 seals
- Confidence grades A-D, computed by displayed rules
- States, hypotheses and the care-plan proposal, computed by the engine from the record
- The seven-day indication, reading a live Open-Meteo forecast for Toulouse
- The co-signature and the change from "proposed" to "validated"
- Replies to citizens, the stream's voice (grounded sentence by sentence), the HL7 FHIR R4 export — accepted by the public HAPI FHIR server

Demonstration data

- People (Mathis, Léa, Inès, Camille...) and their past check-ups; the five-city Stream Cup numbers
- Dr Ferreira, referent ecologist, is a role-play: no real ecologist has signed anything
- The emergency scene: scripted citizens and a scripted heatwave in accelerated time (the engine is real)
- Photos are real urban streams from Wikimedia Commons, used as stand-ins; stream states are not claims about those places
- Background measures (temperature, conductivity) shown in timelines

What we do not claim

- No integration with OneAquaHealth systems: StreamChart is **designed to plug into** the Citizen Science App, the Decision Support System and DipteraCAST, and compatible with public guidelines.
- No diagnosis and no epidemiological forecast: hypotheses carry a confidence; the health indication is indicative, explained and dated.
- Grades are a proposed method, not a validated one. The AI can be wrong on a stream photo — and says "unsure" when it is.
- No official relationship with any city or school.

Sources

- Care-plan measures: OneAquaHealth D2.4 — Catalogue of measures for urban aquatic ecosystems rehabilitation (University of Coimbra et al., 2026).
- Indicators: OneAquaHealth citizen-science indicators judgeable without equipment (foam, colour, odour, flow, banks, riparian vegetation, litter).
- Weather: Open-Meteo forecast API (CC BY 4.0). AI: Google Gemini API, bring-your-own key.
- Standards: HL7 FHIR R4 (Patient, Location, QuestionnaireResponse, Observation, Media, Device, DiagnosticReport, RiskAssessment, CarePlan, Provenance, Communication).

Photo credits (Wikimedia Commons)

- "[Passage du ruisseau de Maltempes sous le canal latéral](#)" — Wikimedia Commons contributor, CC BY-SA 4.0, Toulouse, France
- "[Marcaïssonne](#)" — Adrianstork, CC BY-SA 4.0, Toulouse, France

Language level

Child

Curious

Expert

Light

Dark

Honesty